

FILIPPO GIOVAGNINI

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SUMMARY

PhD candidate in Mathematics at Imperial College London with experience in stochastic analysis, probabilistic modelling, Monte Carlo methods, and machine learning on noisy high-dimensional data.

EXPERIENCE

Stripe Partners

Data Scientist (Part-time)

London, UK

Mar 2026 – Present

- Built embedding-based and statistical modelling pipelines to extract structured features from noisy behavioural and text data.
- Contributed to internal AI research and client-facing analysis by transforming unstructured datasets into model-ready signals.

Aleta Index

ML Engineer (Part-time)

London, UK

Feb 2025 – Feb 2026

- Built real-time quantitative models tracking topic, sentiment, and framing dynamics across heterogeneous financial news sources.
- Designed pipelines converting unstructured financial text into structured features for downstream signal analysis and monitoring.

Imperial Business School

Teaching Assistant — Stochastic Calculus for Finance (MSc)

London, UK

2023 – Present

EDUCATION

Imperial College London

PhD in Mathematics — Stochastic Analysis, MFC CDT

London, UK

Dec 2023 – Present

ETH Zürich

Exchange Program (SEMP) — Quantitative Risk Management, Optimal Transport

Zürich, Switzerland

2023

University of Pisa

MSc in Mathematics — 110/110 cum laude (GPA 30/30)

Pisa, Italy

2023

University of Pisa

BSc in Mathematics — 110/110 cum laude (GPA 29.85/30)

Pisa, Italy

2021

RESEARCH

Research focus: mean-field limits, stochastic PDEs, interacting particle systems, and high-dimensional stochastic dynamics.

Two preprints on particle approximations for fluid-related stochastic PDEs: [arXiv:2410.23163](#) and [arXiv:2504.12960](#).

Recent preprint on universality in deep neural networks: [arXiv:2605.02771](#).

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, Polars, Scikit-learn, PyTorch), C++, MATLAB.

Quantitative: Probability, stochastic calculus, Monte Carlo methods, numerical analysis, probabilistic modelling, statistical learning.

ML / Data: NLP, embedding-based methods, feature engineering, modelling on unstructured data.

Languages: Italian (native), English (fluent), Spanish (conversational).